GymSync: A Smart Gym Management System

An integrated platform built for modern fitness operations — handling check-ins, scheduling, and sales seamlessly, whether online or offline.

CPT 264 · SYSTEMS & PROCEDURES

Made with GAMMA

The Problem with Today's Gym Systems

Most gyms rely on disconnected tools that break down the moment connectivity fails — creating friction for members and staff alike.

Disconnected Systems

Check-ins, sales, and scheduling operate in silos, creating data gaps and manual workarounds.

Internet Dependency

A single outage halts operations entirely — no check-ins, no sales, no record of activity.

Manual Errors

Staff compensate for system gaps with paper logs and spreadsheets, introducing costly mistakes.

Made with GAMMA

Introducing GymSync

GymSync is a unified gym management platform designed to keep operations running — no matter the network conditions.

Member Check-Ins

Fast, frictionless entry with real-time or offline validation.

Class Scheduling

Manage sessions, enrollments, and staff assignments in one place.

Point of Sale

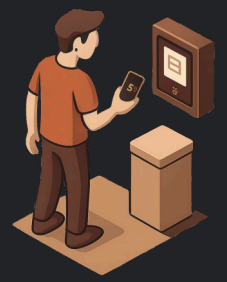
Process memberships, retail, and add-ons with full transaction history.

Offline-First Design

Local data storage ensures zero downtime during internet outages.

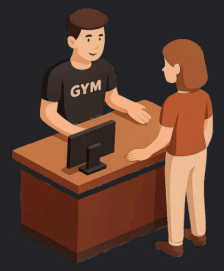
Made with GAMMA

Who GymSync Serves



Members

Quick check-ins, easy class bookings, and a smooth experience every visit.



Staff

Streamlined workflows for daily operations — check-ins, sales, and scheduling at their fingertips.

Managers

Real-time visibility into attendance, revenue, and system health across all locations.

Made with GAMMA

Project Scope

✅ In Scope

Core operational features that GymSync is built to handle from day one.

- → Member Check-Ins
- → Class Scheduling
- → Point of Sale Transactions

🚫 Out of Scope

These areas are intentionally excluded to keep the system focused and maintainable.

- Payroll Processing
- Full Accounting
- Advanced Analytics

Made with GAMMA

Key System Requirements

Functional Requirements

- Member check-in with online/offline support
- POS transaction processing
- Class scheduling and enrollment
- Offline data capture and sync

Non-Functional Requirements

- **Reliability** — system stays operational under all conditions
- **Security** — member and transaction data protected
- **Performance** — fast response times at peak hours
- **Data Sync** — accurate reconciliation after outages

Made with GAMMA

Core Use Cases

GymSync covers four primary workflows that define the daily operations of any fitness facility.

1

Member Check-In

Validate membership and grant facility access — online or offline.

2

Process Sale (POS)

Handle memberships, retail items, and add-on services with full receipt tracking.

3

Manage Class Scheduling

Create, modify, and assign staff to classes; manage member enrollments.

4

Sync Offline Data

Automatically reconcile queued transactions when connectivity is restored.

Bonus: GymSync also generates sales and attendance reports for management oversight.

Made with GAMMA

Process Model & Data Architecture

Process Flow Highlights

Every operation follows a decision-based workflow that branches on connectivity status, ensuring no action is ever lost.

- Check-in flow with online/offline routing
- POS transaction process with local fallback
- Scheduling workflow with conflict detection
- **Pending Sync Queue** — guarantees zero data loss during outages

Data Model Entities

Member

Profiles, status, and history

Transaction

Sales, payments, and receipts

Class

Schedule, capacity, and enrollment

Staff

Roles, assignments, and access

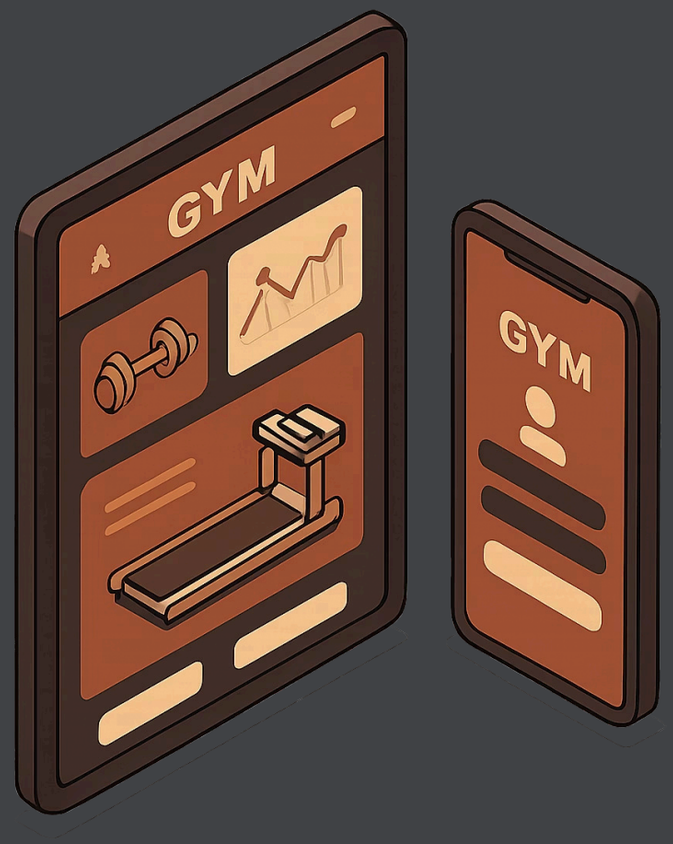
Made with GAMMA

System Architecture

GymSync uses an **N-tier Client-Server architecture** with a cloud-primary design and a local fallback layer for uninterrupted operations.

The sync engine is the backbone of GymSync's reliability — it automatically detects reconnection and pushes all queued data to the cloud, ensuring a single source of truth at all times.

Made with GAMMA



User Interface & Implementation

UI Design Focus

Three core screens — Check-In, POS, and Scheduling Dashboard — designed for speed, simplicity, and clear online/offline status indicators.

Program Logic

Decision-based workflows with online/offline branching ensure consistent behavior across all network conditions.

Data Storage

Dual-layer storage: cloud database for primary operations, local database for offline resilience.

Implementation Status

A front-end prototype simulates full system behavior, validating workflows before production deployment.

Conclusion: GymSync is a reliable, scalable solution purpose-built for the demands of modern gym operations.

Made with GAMMA